

Phase 1: Real Word Targets & Statistics

Target Selection: I started by generating and selecting a specific list of real English words to serve as my experimental targets.

CLEARPOND Statistics: I ran these real words through the CLEARPOND database to extract their baseline phonological and orthographic statistics. Specifically, I extracted the exact Word Length, Number of Phonological Neighbors (NPN), and Phonological Token Average Frequency (PTAF) for every real word.

Phase 2: Sourcing the Nonword Pool

The ARC Database: To find matching nonwords, I utilized the Macquarie University ARC Nonword Database (Rastle, Harrington, & Coltheart, 2002), which contains over 350,000 pre-generated, phonotactically legal nonwords and their pre-calculated neighborhood statistics.

PTAF Correction: The raw ARC Database provides the Summed Frequency of Phonological Neighbors (SFPN), but I needed the Average (PTAF) to match my CLEARPOND data. Using a custom Python script, I mathematically calculated the PTAF for every nonword in the ARC database using the formula: $PTAF = SFPN / NPN$.

Phase 3: Filtering the Nonwords

To ensure the nonwords were high-quality and unambiguous to read, my script aggressively filtered the massive ARC database, removing any problematic nonwords. The exclusion criteria removed: 5. Ambiguous Spellings: Nonwords containing tricky consonant clusters (like ph, kn, gh, ps) or double consonants (like bb, tt). 6. Grammatical Endings: Nonwords ending in -s, -ed, or -e, preventing participants from confusing them with plurals, past-tense verbs, or words with a "silent e". 7. Real-Word Overlap: Finally, my script checked every nonword against a standard English dictionary. If a nonword could be turned into a real English word by changing just a single letter, it was completely excluded.

Phase 4: Algorithmic Matching

Finding the Perfect Match: With a clean pool of highly constrained ARC nonwords, my algorithm was executed. For every single real word I had, the script searched the remaining ARC pool and selected exactly one nonword that:

Exactly matched the character length of the real word.

Was mathematically closest to the real word's NPN and PTAF (based on normalized distance).

(Methodological Note: Target real word properties were extracted from the CLEARPOND database, while matched nonwords and their corresponding properties were sourced from the ARC Nonword Database.)

To ensure each nonword was optimally matched to its real-word target across two fundamentally different metrics—Number of Phonological Neighbors (NPN) and Phonological Token Average Frequency (PTAF)—a normalized distance scoring algorithm was implemented. First, because the ARC database natively reports the Summed Frequency of Phonological

Neighbors (SFPN) rather than an average, PTAF was mathematically derived for all nonword candidates using the formula:

$$PTAF = \frac{SFPN}{NPN}$$

PTAF = Phonological Token Average Frequency

SFPN = Sum Frequency of Phonological Neighbors

NPN = Number of Phonological Neighbors

Next, because NPN and PTAF exist on vastly different numerical scales, a raw difference score could not be used to find the closest match. Instead, the absolute difference between the target word and the candidate nonword was calculated for both metrics, and each difference was divided by the standard deviation of that metric across the entire target word set. These two normalized values were then summed to produce a single, unified discrepancy score for every candidate. For each real word, the nonword with the lowest total discrepancy score—representing the shortest mathematical distance across both metrics simultaneously—was algorithmically selected. While target statistics were sourced from the CLEARPOND database and candidate statistics from the ARC database, this normalization procedure ensured that the selected nonwords represented the closest possible relative matches available in the candidate pool.